

Vitamin	Function	Sources
B1 (Thiamine)	Vitamin B1, commonly referred to as thiamine, is critical in the metabolism of sugar and amino acids as well as central nervous system function. Research on the effect of vitamin B1 supplementation on performance in humans is equivocal and the effects do not appear to overly robust.	• Enriched and fortified grains • Pork • Salmon • Flax seeds • Squash • Black beans • Tuna • Milk • Beef
B2 (Riboflavin)	Vitamin B2, commonly referred to as riboflavin, is critical in the metabolism of proteins, carbohydrates, and fat and is a component of flavin adenine dinucleotide (FAD). FAD is an essential part of the antioxidant system in the body, specifically via its roles in regulating glutathione and uric acid, which are two of the main antioxidants in the body. There does not appear to be any benefit of vitamin B2 supplementation on athletic performance.	• Enriched and fortified grains • Beef • Tofu • Milk • Fish • Mushrooms • Pork • Spinach • Almonds • Avocados • Eggs
B3 (Niacin)	Vitamin B3, commonly referred to as niacin, shares many similarities with vitamin B2, in that is critical in the metabolism of proteins, carbohydrates, and fat and is a component of a dinucleotide, specifically, nicotinamide adenine dinucleotide (NAD). NAD is an important molecule, it plays a critical role in the Krebs's cycle and electron transport chain and is responsible for much of the ATP	• Enriched and fortified grains • Fish • Chicken • Turkey • Pork • Beef • Peanuts • Brown rice • Mushrooms • Avocados

	production in our cells. Furthermore, NAD is involved in the production of many other macromolecules that are essential for optimal function, including cholesterol and fatty acids. There does not appear to be any robust benefit of B3 supplementation on athletic performance.	
B5 (Pantothenic Acid)	Vitamin B5, commonly referred to as pantothenic acid, is a component to co-enzyme A (CoA). CoA serves as a molecular link between glycolysis and the Krebs's cycle, which means that it is involved in the metabolism of proteins, carbohydrates, and fatty acids. Furthermore, it is a fundamental part of the production cholesterol, fatty acids, and other molecules, such as coenzyme Q10 (CoQ10). There does not appear to be any robust benefit of B5 supplementation on athletic performance.	• Enriched and fortified grains • Salmon • Shiitake mushrooms • Chicken • Beef • Milk • Seeds • Sweet potatoes • Lentils
B6	Vitamin B6 represents a class of several related, but different molecules: pyridoxal, pyridoxine, and pyridoxamine. Each of these molecules can be have a 5' linked phosphate (pyridoxal 5'-phosphate, pyridoxine 5'-phosphate, and pyridoxamine 5'-phosphate). Several of these, specifically pyridoxal 5'-phosphate and pyridoxamine 5'-phosphate play critical roles in metabolism. Pyridoxal 5'-phosphate is a coenzyme for glycogen phosphorylase, the enzyme that breaks down glycogen into glucose molecules (specifically	• Enriched and fortified grains • Whey protein • Milk • Cheese • Eggs • Tuna • Organ meats • Potatoes • Bananas • Parsnips • Pistachios

	glucose-1-phosphate) for energy production. Pyridoxal 5'-phosphate is also involved in the production of neurotransmitters (e.g. GABA, dopamine, and serotonin), and gluconeogenesis. Cumulatively, while vitamin B6 does alter fuel metabolism slightly, it does not appear to either impair or improve performance.	
Folate	Similar to vitamin B12, folate is unique among the b-vitamins, however it plays a much narrower role in human metabolism than vitamin B12. Folate helps regulate single-carbon transfers, both accepting and donating single carbons to chemical reactions. Folate supplementation has not been shown to improve athletic performance, and the research surrounding the effect of folate on athletic performance is scarce. However, folate supplementation can be an important part of a nutritional strategy for certain populations. For example, in pregnant women, folate supplementation may improve hemoglobin levels and reduce birth complications. Furthermore, it may also help with recovery from depressive disorders.	• Enriched and fortified grains • Seafood • Lean meats and poultry • Eggs • Legumes (beans and peas) • Nuts • Seeds • Soy products
Vitamin B7 (Biotin)	Vitamin B7, commonly referred to as biotin, and formerly referred to as vitamin H, is a cofactor in many carboxylase enzymes, which are responsible for transferring carbon dioxide molecules and the metabolism of carbohydrates, fatty acids,	• Liver • Organ meats • Eggs (yolk) • Nuts • Seeds • Salmon • Yeast • Milk

	cholesterol, and amino acids. There is no evidence that biotin supplementation has any impact on athletic performance and the research surrounding this is scant. Much of the biotin research as it applies to health has been focused on the benefit on hair, nail, and skin. While there are robust marketing claims to support biotin supplementation as a beauty aide, and one study showing a small benefit on nail health, the cumulative scientific data do not support it.	• Eggs • Cheese • Sweet potatoes • Avocados
B12	Vitamin B12 stands out among the b-complex vitamins in that it is the largest, comes in many forms, and contains a heavy metal/mineral in its molecular structure, specifically cobalt. Vitamin B12 is involved in a panoply of metabolic processes, including DNA synthesis, red blood cell production, and in maintaining proper neurological function. Vitamin B12 cannot be made by the body, and as such, it must be consumed through the diet. While vitamin B12 supplements are often marketed as “energy supplements, there is little evidence examining the effect of vitamin B12 supplementation on athletic performance.	• Enriched and fortified grains • Shellfish • Liver • Trout • Salmon • Tuna • Beef • Milk • Cheese • Yogurt • Eggs
Vitamin C	Vitamin C, commonly referred to as ascorbic acid, plays myriad roles in the human body: 1) it serves as an antioxidant by directly scavenging free radicals and by recycling vitamin E, 2) it is	• Bright colored fruit (e.g. oranges, strawberries, guavas, cherries) • Bell peppers • Cruciferous vegetables • Dark leafy greens

	involved in the production of carnitine, which shuttles fatty acids into the mitochondria, and 3) it is essential to produce collagen, which gives bones, vessels, tendons, and ligaments their physical properties. The current status of the evidence is mixed, and it is not entirely certain whether vitamin C supplementation is beneficial for exercise or detrimental, and in what circumstances it might play either role.	• Tomatoes
Choline	Choline is not considered a vitamin but is a water-soluble essential micronutrient. Choline plays many important roles, but the two main roles of choline is to aide in providing structure for cell membranes and as a critical molecule to produce the neurotransmitter acetylcholine. It also plays a role in regulating metabolism of methyl groups, specifically in homocysteine. The body manufactures choline in the liver but does not make enough to cover all the choline needs of the body, meaning that choline must be consumed in the diet. Currently, the data do not provide evidence to support the use of choline as an ergogenic aid.	• Liver • Eggs • Oysters • Mushrooms • Cauliflower • Dark leafy greens